



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO DEPT. OF SUGAR - DHARMAPURI DCSM LT

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

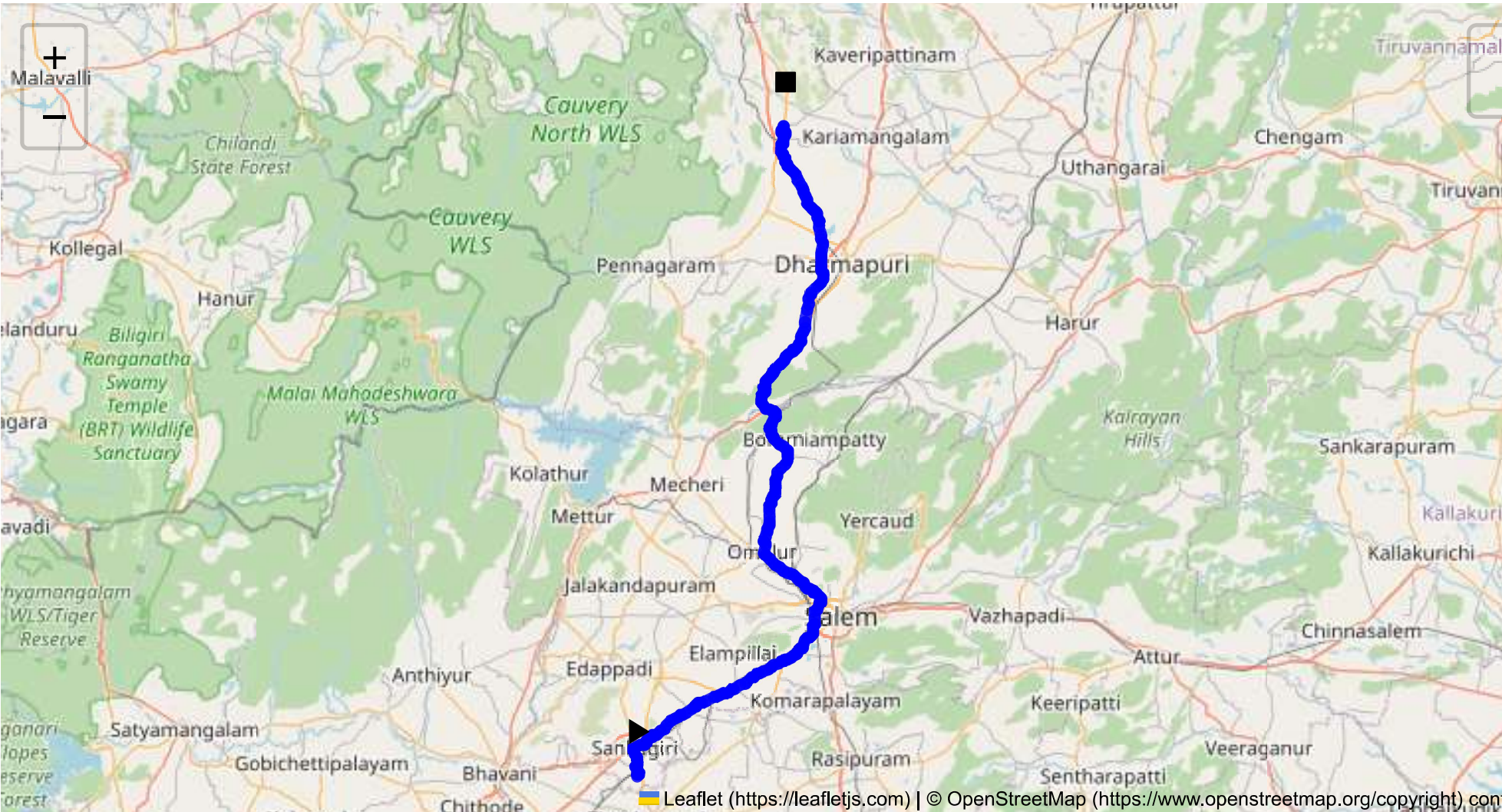
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 127.88 km**  
**Estimated Duration: 2.4 hours**  
**Adjusted Duration (Heavy Vehicle): 3.0 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.322287, 78.076318)**



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Route Overview

The route from Sangagiri to Palacode covers approximately 127.88 kilometers. It primarily traverses through NH44, with sections on state highways and local roads leading into Palacode. The journey



typically takes around 2.41 hours for trucks carrying hazardous materials.

## 2. Weather Conditions

- **Typical Weather:** The area generally experiences tropical weather. Summers can be extremely hot, while monsoons bring heavy rains.
- **Potential Hazards:** Monsoon season often leads to flooding and slippery roads. Visibility issues due to fog can occur during winter mornings.

## 3. Traffic Patterns

- **Peak Hours:** Between 7:30-10:00 AM and 5:00-8:00 PM. Significant congestion is noted in towns like Dharmapuri.
- **Congestion-Prone Areas:** Urban centers and toll booths along NH44 often experience delays.

## 4. Road Quality and Infrastructure

- **NH44:** Well-maintained with four-lane sections; however, occasional roadworks might impede smooth movement.
- **State Highways:** Mixed conditions. Potholes are occasional, especially post-monsoon.
- **Local Roads to Palacode:** Narrower and occasionally under-maintained, requiring cautious driving.

## 5. Alternative Routes

- **Emergency Detour:** Via Erode and Sivaganga for avoiding NH44; significantly longer but useful in the event of roadblocks or closures on the primary route.

## 6. Local Regulations

- **Hazardous Material Transport:** Restricted during peak hours in urban areas and disallowed in environments with dense residential settlements.
- **Permits:** Required for transporting specific hazardous materials across state lines.

## 7. Historical Incidents

- Notable accidents involve vehicle collisions and rollovers, often due to overspeeding or vehicle overloading. Minimal incidents are directly linked to hazardous materials.

## 8. Environmental Considerations

- **Sensitive Areas:** Pollution control zones around agricultural lands and water bodies.
- **Protected Areas:** Drivers must be mindful near forested regions closer to Dharmapuri.

## 9. Communication Coverage

- **Reliable Signal:** Predominantly along NH44.
- **Dead Zones:** Some patchy coverage in hilly sections and rural areas approaching Palacode.

10. Emergency Response Times

- **Urban Sections:** 15-30 minutes due to proximity to service centers.
- **Rural Areas:** Up to 45-60 minutes, especially if connectivity issues arise.

11. Overall Risk Assessment

- **Medium Risk:** Combination of effective infrastructure on NH44 with challenges posed by local road conditions and weather variability.
- **Mitigation:** Ensure compliant speed, remain updated with weather forecasts, and maintain open communication with control centers for any anomalies.

Summary

This route, though mostly reliable, requires strategic navigation around peak times and adverse weather conditions. Drivers should maintain vigilance regarding road quality issues, primarily on secondary and tertiary roads. Emergency preparedness is critical, with routine checks on vehicle status and communication devices to mitigate any potential risks en route.

Risk Assessment - Turns

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 0  | Turn      | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2  | Turn      | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3  | Turn      | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 4  | Turn      | Medium     | 11.45309, 77.87325 | 30 KM/Hr    |
| 5  | Turn      | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 6  | Turn      | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 7  | Turn      | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 8  | Turn      | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 9  | Turn      | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 10 | Turn      | Medium     | 11.53501, 77.95707 | 30 KM/Hr    |
| 11 | Turn      | Medium     | 11.53601, 77.95856 | 30 KM/Hr    |
| 12 | Turn      | Medium     | 11.64180, 78.11808 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 13 | Turn      | Medium     | 11.64325, 78.11993 | 30 KM/Hr    |
| 14 | Turn      | Medium     | 11.69519, 78.10775 | 30 KM/Hr    |
| 15 | Turn      | Medium     | 11.73958, 78.05105 | 30 KM/Hr    |
| 16 | Turn      | Medium     | 11.75835, 78.05016 | 30 KM/Hr    |
| 17 | Turn      | Medium     | 11.86776, 78.08342 | 30 KM/Hr    |
| 18 | Turn      | Medium     | 11.87007, 78.08384 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.91345, 78.05865 | 30 KM/Hr    |
| 20 | Turn      | Medium     | 11.92545, 78.06496 | 30 KM/Hr    |
| 21 | Turn      | Medium     | 11.93691, 78.05433 | 30 KM/Hr    |
| 22 | Turn      | Medium     | 11.95498, 78.04689 | 30 KM/Hr    |
| 23 | Turn      | High       | 11.95847, 78.04989 | 15 KM/Hr    |
| 24 | Turn      | Medium     | 11.96062, 78.04957 | 30 KM/Hr    |
| 25 | Turn      | Medium     | 11.96366, 78.05142 | 30 KM/Hr    |
| 26 | Turn      | Medium     | 11.96427, 78.05118 | 30 KM/Hr    |
| 27 | Turn      | High       | 11.96526, 78.04950 | 15 KM/Hr    |
| 28 | Turn      | High       | 11.96637, 78.04923 | 15 KM/Hr    |
| 29 | Turn      | Medium     | 11.96859, 78.05102 | 30 KM/Hr    |
| 30 | Turn      | Medium     | 12.12027, 78.13284 | 30 KM/Hr    |
| 31 | Turn      | High       | 12.26533, 78.08428 | 15 KM/Hr    |
| 32 | Turn      | High       | 12.26550, 78.08440 | 15 KM/Hr    |
| 33 | Turn      | Medium     | 12.27213, 78.08087 | 30 KM/Hr    |
| 34 | Turn      | Medium     | 12.27348, 78.08085 | 30 KM/Hr    |
| 35 | Turn      | High       | 12.29595, 78.07368 | 15 KM/Hr    |
| 36 | Turn      | High       | 12.29651, 78.07545 | 15 KM/Hr    |
| 37 | Turn      | High       | 12.31195, 78.07847 | 15 KM/Hr    |
| 38 | Turn      | High       | 12.31360, 78.07464 | 15 KM/Hr    |

Emergency Locations

|   | type     | name              | coordinates            | speed_limit | risk_level |
|---|----------|-------------------|------------------------|-------------|------------|
| 0 | hospital | Thilagam Hospital | 11.4753573, 77.8685829 | 30 km/h     | Medium     |

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 1  | clinic   | Meyyazhagan Clinic                                              | 11.4752254,<br>77.8685671 | 30 km/h     | Medium     |
| 2  | hospital | Dr. Jaganathan Hospital                                         | 11.4752717,<br>77.8709312 | 30 km/h     | Medium     |
| 3  | hospital | Kathir Eye Hospital                                             | 11.4740976,<br>77.8681591 | 30 km/h     | Medium     |
| 4  | police   | Police Quarters                                                 | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5  | hospital | Sankari Hospital                                                | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 6  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 7  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 9  | hospital | C. N. Hospital                                                  | 11.55724,<br>78.0096      | 30 km/h     | Medium     |
| 10 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 12 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 19 | hospital | Bharathi Hospital, Salem                                        | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 20 | hospital | Sisu Hospital Inida Ltd                                         | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 21 | hospital | Aishwarya Hospital                                              | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 22 | hospital | C.M. Hospital                                                   | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |
| 24 | pharmacy | Healthy Pharmacy                                                | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 25 | pharmacy | Thulasi Pharmacy                                                | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 26 | hospital | Sri Gokulam Speciality Hospital                                 | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 27 | hospital | Global Medical Center                                           | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 28 | hospital | Manipal Hospital                   | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |
| 30 | pharmacy | MJ Medicals                        | 11.7232728,<br>78.0698965 | 30 km/h     | Medium     |
| 31 | pharmacy | Salem Medicals                     | 11.7343192,<br>78.0566844 | 30 km/h     | Medium     |
| 33 | clinic   | Manimuthu Skin Clinic              | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |
| 34 | hospital | St. Joseph's Speciality Hospital   | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |
| 36 | hospital | Anugraha Multi Speciality Hospital | 11.773549,<br>78.0546408  | 30 km/h     | Medium     |
| 38 | hospital | L.R. Hospital                      | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |
| 39 | clinic   | JSA Clinic                         | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 40 | hospital | Thoppur,Government Hospital        | 11.9398824,<br>78.0551658 | 30 km/h     | Medium     |
| 42 | hospital | Government Hospital Palacode       | 12.2964824,<br>78.0761735 | 30 km/h     | Medium     |

Crowded Spots

|    | type       | name                                                       | coordinates               | speed_limit | risk_level |
|----|------------|------------------------------------------------------------|---------------------------|-------------|------------|
| 13 | university | Vinayaka Missions Vinayaka Missions<br>Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 14 | school     | Vinayaka Missions Annapoorani Matric<br>School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 16 | school     | Dist Education Training                                    | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 17 | school     | DIET - Salem                                               | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 18 | school     | Diamond Rays Matric HR Sec School                          | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 23 | college    | Sona College                                               | 11.6768,<br>78.1257052    | 30 km/h     | Medium     |



|    | type       | name                                                 | coordinates               | speed_limit | risk_level |
|----|------------|------------------------------------------------------|---------------------------|-------------|------------|
| 29 | university | Periyar University                                   | 11.7186758,<br>78.0771089 | 30 km/h     | Medium     |
| 32 | school     | Fathima Hr Sec School                                | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 35 | school     | John Britto Matriculation Higher<br>Secondary School | 11.7676907,<br>78.0513092 | 30 km/h     | Medium     |
| 37 | college    | Dhirajlal Gandhi College of Technology               | 11.7968014,<br>78.0562095 | 30 km/h     | Medium     |
| 41 | college    | Laxminarayana Polytechnic College                    | 12.0421277,<br>78.103203  | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44900, 77.87406



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45028, 77.87202



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45309, 77.87325





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46400, 77.87323



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46765, 77.87037





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47403, 77.86841



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49149, 77.89601



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49224, 77.89601





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53501, 77.95707



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53601, 77.95856





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64180, 78.11808



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64325, 78.11993



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.69519, 78.10775





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73958, 78.05105



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.75835, 78.05016





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86776, 78.08342



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87007, 78.08384



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.91345, 78.05865





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92545, 78.06496



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93691, 78.05433





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95498, 78.04689



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95847, 78.04989



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96062, 78.04957





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96366, 78.05142



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96427, 78.05118





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96526, 78.04950



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96637, 78.04923



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96859, 78.05102





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12027, 78.13284



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.26533, 78.08428





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.26550, 78.08440



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.27213, 78.08087



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.27348, 78.08085





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.29595, 78.07368



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.29651, 78.07545





Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.31195, 78.07847



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.31360, 78.07464